



DPUSCNXForesight

Calculation of road geometry at a position in front or behind the EGO vehicle.

1. Dynamic parameters:

Distance

Foresight distance [m] to the reference point defined by the ManagedCursorIndex. Positive values specify positions in front and negative values specify positions behind the reference point.

Heading

Expected driving directions of the vehicle. (default: 1 = STRAIGHT; 2 = RIGHT; 4 = LEFT.)

2. Static parameters:

ManagedCursorIndex

ID of the cursor which is used as reference point for calculation of the foresight. (default = 0: primary cursor providing the position of the EGO vehicle).

3. Outputs:

X, Y, Z

Coordinates of the preview point measured in the simulation coordinate system. A closer description of this coordinate system can be found in „Reference Visualization SGE“.

ZOffset

Height difference between reference point and foresight point.
 $Z_{\text{Offset}} = Z_{\text{foresight}} - Z_{\text{ref}}$.

4. Dependencies to other DPUs:

DPUSCNXForesight depends on functions of **DPUSCNX**, which must run on the computer(s) on which it is used. The Index value of **DPUSCNXForesight** must be higher than the Index value of **DPUSCNX**. Additionally the **DPUSCNXForesight** has to run in the same Thread as **DPUSCNX**.

5. Example:

The DPUSCNXForesight can be used as follows:

```
DPUSCNXForesight InclineWarning
{
    Computer = {SCN};
    Index = 35;
    Distance = 200; # Foresight distance
};
Connections = {
    InclineWarning.ZOffset -> ...
};
```